

Ishaan Agarwal

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EDUCATION

Cornell University, Cornell Ann S. Bowers College of Computing and Information Science

Ithaca, NY

Master of Professional Studies, Applied Statistics, **GPA: 3.87/4.0**

May 2026

Selected Coursework: Modern Regression, Categorical Data, Data Mining & Machine Learning, Statistical Computing – I, Applied Time Series Analysis, Bayesian Data Analysis, Applied Neural Networks, Introduction to Deep Learning

Manipal Institute of Technology

Karnataka, India

Bachelor of Technology in Electronics and Communication Engineering

June 2023

Minor concentration: Data Science

Selected Coursework: Data Structures & Algorithms, Database Systems, Regression Models, Statistical Inference, Machine Learning, Linux Programming, Engineering Mathematics I–IV

SPECIALIZED SKILLS

Languages & Libraries: Python, R, SQL, Java, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, HuggingFace, Spring Boot

Cloud, Data & Tools: Azure, Kubernetes, Git, GitLab CI/CD, Docker, Kafka, REST APIs, Redis, PostgreSQL, Google Cloud

Methods: Statistical Analysis, Time Series Forecasting, Causal Inference, Machine Learning, Computer Vision, RAG, LLMs

Certifications: Microsoft Certified: Azure AI Fundamentals (AI-900), Deep Learning Specialization (DeepLearning.AI)

EXPERIENCE

Software Engineer – Data & Search Infrastructure, UBS Bank, India

Full-Time following internship

January 2023 - July 2025

- Managed and extended 20+ Azure Search indexes powering UBS Neo's cross-asset trading platform (1.8M+ users), ingesting real-time data via Kafka, Tibco, and REST APIs to support 200K daily API calls.
- Built a RAG pipeline prototype to automate tagging of 10,000+ financial documents using LLM-generated tags from similar document context, reducing manual effort by 60%; Top 5 in UBS AI Quest Competition.
- Led the end-to-end migration of a business-critical search service to a modern cloud-native AKS architecture, eliminating redundant service-level authentication in favor of platform-wide verification, delivered with zero production incidents.
- Implemented region-based data segmentation for the FX search pipeline by extending the Azure Search schema, executing a full reindex of 2.4M+ financial instrument records.
- Migrated a core search service from 29West to gRPC, eliminating .dll dependencies to enable AKS deployment; migrated services to Azure Cache for Redis, maintaining a real-time data quality monitoring pipeline tracking daily ingestion counts across upstream sources.

Data Science Intern, Cogoport, India

June 2021 - July 2021

- Developed LSTM models for freight rate forecasting, achieving RMSE of ~\$1,200 against an average freight price of ~\$7,000; engineered geospatial features by clustering shipping ports to capture route-level patterns, outperforming baseline ARIMA models.

RESEARCH

Causal ML: Embedding-Based Adjustment for Household Economics

- Investigating whether LLM embeddings (MiniLM, DistilBERT) of household histories can serve as valid proxies for latent confounders in causal models, supervised by Prof. Jelena Bradic at Cornell University.
- Estimating the Average Treatment Effect (ATE) of job-related training on earnings using the Panel Study of Income Dynamics (PSID), applying Double Machine Learning (DML) with cross-fitting to control for hidden confounders in observational panel data.

RELEVANT PROJECTS

Inductive Bias & Sample Efficiency: CNNs vs Vision Transformers

- Benchmarked a lightweight ResNet against a Minimal Vision Transformer (ViT) on progressive data splits (1%–100%) to explicitly quantify the impact of spatial inductive bias on model sample efficiency.
- Demonstrated the CNN's superior sample efficiency, achieving 84.55% test accuracy with only 5% of training data, whereas the ViT required 100% data scale and augmentation to hit a 95.66% peak.

High-Dimensional Gene Network Analysis

- Estimated sparse precision matrices from high-dimensional breast cancer datasets ($p \gg n$) by implementing Constrained L1-Minimization (CLIME) and Graphical Lasso.
- Reconstructed gene regulatory networks to predict clinical responses via Linear Discriminant Analysis (LDA), benchmarking the convergence rates of convex (CLIME) against non-convex (SCAD) penalization methods.